



Barker
College

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Student Number

2023

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Advanced

Staff Involved:

- DXC - D. Chua
- JAI - J. Iles
- LAK - L. Kalnins
- PDJ - P. Jouliany

- LMD - L. de Gorter
- DZB - D. Batchen
- ARM - A. Mallam
- ARP - A. Perkins

Friday, August 4th 2023

150 copies

- RJW - R. Williams
- ALY - A. Young
- RAS - R. Smith*

General

Instructions:

- Reading time - 10 minutes
- Writing time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:

100

Section I - 10 marks (pages 2 - 6)

- Attempt Multiple Choice Questions 1 - 10
- Allow about 20 minutes for this section

Section II - 90 marks (pages 7 - 28)

- Attempt Questions 11 - 37
- Show all necessary working
- Allow about 2 hours and 40 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 20 minutes for this section

Use the Multiple-Choice answer sheet for Questions 1-10.

1. What is the solution to the equation $2 \cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$?

- A. $x = \frac{\pi}{3}$ and $\frac{5\pi}{3}$
- B. $x = \frac{\pi}{3}$ and $\frac{2\pi}{3}$
- C. $x = \frac{\pi}{6}$ and $\frac{5\pi}{6}$
- D. $x = \frac{\pi}{6}$ and $\frac{11\pi}{6}$

2. What is the range of the function $y = 3^{-x} + 2$?

- A. $(3, \infty)$
- B. $(-\infty, 2)$
- C. $(2, \infty)$
- D. $(-\infty, 1)$

3. What is $\int e + e^{4x} dx$?

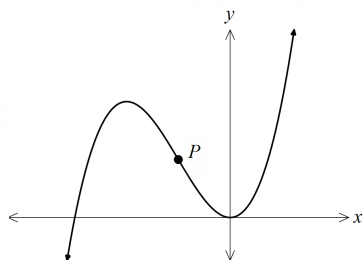
- A. $ex + 4e^{4x} + C$
- B. $ex + \frac{1}{4}e^{4x} + C$
- C. $e + 4e^{4x} + C$
- D. $e + \frac{1}{4}e^{4x} + C$

4. There are 15 ties in a cupboard. 8 of the ties are silk and 9 of the ties are striped. 3 ties are neither silk nor striped.

If one tie is selected at random, what is the probability that it is silk but not striped?

- A. $\frac{8}{17}$
 B. $\frac{4}{15}$
 C. $\frac{1}{5}$
 D. $\frac{2}{17}$

5. Which of the following statements are true about the point of inflection, P , on the curve below?



- A. $f'(x) < 0, f''(x) = 0$
 B. $f'(x) < 0, f''(x) > 0$
 C. $f'(x) > 0, f''(x) = 0$
 D. $f'(x) > 0, f''(x) < 0$

6. The probability distribution of a random variable X is shown below:

x	0	1	2	3	4
$P(X = x)$	$\frac{1}{5}$	k	$2k$	k	$\frac{2}{5}$

What is the value of k ?

- A. $\frac{1}{4}$
 B. $\frac{3}{5}$
 C. $\frac{3}{20}$
 D. $\frac{1}{10}$

7. An infinite geometric series has a first term of 12 and a limiting sum of 18. What is the common ratio?

- A. $\frac{1}{3}$
 B. $\frac{1}{2}$
 C. $\frac{2}{3}$
 D. $\frac{3}{4}$

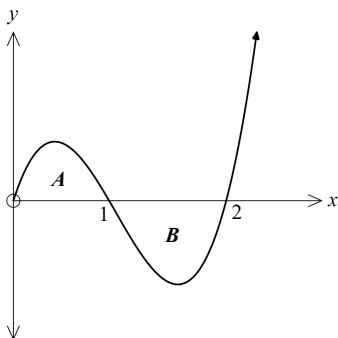
8. What is the period of the function $f(x) = -3\cos\left(\frac{\pi x}{5}\right)$?

- A. $\frac{1}{5}$
- B. $\frac{\pi}{5}$
- C. 10
- D. 10π

9. The function $g(x)$ is even and part of the graph $y = g(x)$ is shown below.

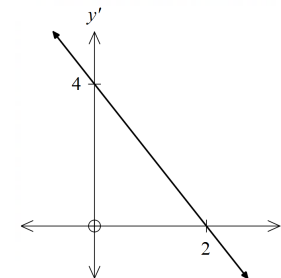
It is known that the area of region A is 3 units² and that $\int_{-2}^2 g(x) dx = -2$.

What is the area bound by the function $y = g(x)$ and the x -axis from $x = -2$ to $x = 2$?



- A. 2 units²
- B. 8 units²
- C. 14 units²
- D. 16 units²

10. The graph of $y' = f'(x)$ is shown below.



The curve $y = f(x)$ has a maximum value of 20. What is the equation of the curve?

- A. $y = x^2 - 4x + 16$
- B. $y = x^2 - 4x + 20$
- C. $y = 20 + 4x - x^2$
- D. $y = 16 + 4x - x^2$

End of Section I

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Section II

90 marks
 Attempt Questions 11 – 37
 Allow about 2 hours and 40 minutes for this section.
 Write your student number on all answer pages.

In Questions 11 – 37, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (1 mark)

Differentiate $\frac{\sin x}{x}$ 1

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Question 12 (3 marks)

Given that $g(x) = 2x + 1$ and $h(x) = x^2 - x$, find:

- (a) $h(g(-2))$ 1
-
-
-
- (b) $g(h(x))$ 1
-
-
-
- (c) x when $h(x) = 0$ 1
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Question 13 (2 marks)

A biologist found that the frequency F (in beats per minute) that a bird beats its wings is inversely proportional to the length L (in cm) of its wings.
 He observes that a bird with wings of length 14 cm beats them at a frequency of 9 beats per minute.

Find an equation relating F and L . 2

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Question 14 (2 marks)

The first term of an arithmetic series is 34. The sum of the first 14 terms is 2023.
 What is the common difference of the series? 2

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Question 15 (2 marks)

Evaluate $\int_0^{\frac{\pi}{6}} \sec^2 2x \, dx$ 2

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Question 16 (3 marks)

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Find the equation of the normal to the curve $y = 2(4x - 3)^4$ at the point on the curve where $x = 1$. **Leave your answer in general form.** **3**

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Question 17 (3 marks)

$$\int_1^7 \frac{8}{x+1} dx$$

Use the trapezoidal rule with 3 sub-intervals to approximate the above integral. Leave your answer in exact form. **3**

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Question 18 (2 marks)

Barry and Doreen are taking a driving test. There is a 70% chance of Barry passing his test and there is an 80% chance of Doreen passing her test. What is the probability that at least one of them passes their test? **2**

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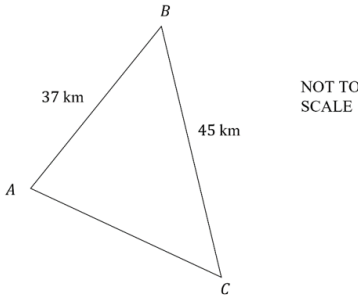
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Question 19 (3 marks)

There are three towns in a regional area. Town B is 37 km from Town A on a bearing of 050°T . Town C is 45 km from Town B on a bearing of 165°T .



- (i) Find the size of $\angle ABC$. **1**

- (ii) Find the distance from Town A to Town C . Round your answer to 1 decimal place. **2**

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Question 20 (2 marks)

Student Number

A Barker student visits *Sultan's Kebab House* every afternoon after school.
 Let the discrete random variable X be the number of kebabs they order on a particular afternoon.

x	0	1	2	3
$P(X = x)$	0.1	0.35	0.4	0.15

- (i) Calculate $E(X)$. 1
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- (ii) The kebab house has a deal for loyal customers.
- Once you purchase a cumulative total of 24 kebabs, you get an exclusive *Sultan's* hoodie.
- After how many visits would a student expect to have ordered enough kebabs to win the hoodie? 1
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Question 21 (5 marks)

The population, P , of penguins in an island colony is decreasing at a rate proportional to P :

$$\frac{dP}{dt} = -kP \quad \text{where } t \text{ is the time in years since 2013.}$$

In 2013 there was a population of 5000 penguins. By 2018 the population was 4250.

- (i) Show that $P = Ae^{-kt}$ is a solution to $\frac{dP}{dt} = -kP$. 1
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- (ii) Find the value of k , rounding your answer to four decimal places. 2
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- (iii) What is the rate of change of the penguin population in 2023? 2
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Question 22 (2 marks)

Consider the following function:

$$f(x) = \begin{cases} x^2, & x < 0 \\ 2x - 3, & x \geq 0 \end{cases}$$

- Find the value of x if $f(x) = 25$. 2
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Question 23 (3 marks)

The graph of the function $f(x) = x^2$ is translated a units to the left, dilated vertically by a factor of b and then translated 3 units up.

The equation of the transformed function is now $g(x) = 2x^2 + 16x + 35$.

Find the values of a and b . 3

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Question 24 (2 marks)

A probability density function is defined as $f(x) = \frac{4}{81}(4x^2 - x^3)$ for $1 \leq x \leq 4$.
What is the mode of this function? 2

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Question 25 (3 marks)

(i) Find $\frac{d}{dx} 2x \ln x$. 1

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(ii) Hence, or otherwise, find $\int 1 + \ln x \, dx$. 2

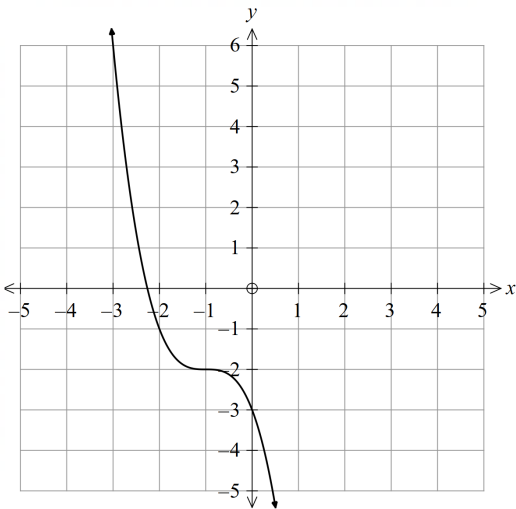
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Question 26 (2 marks)

Consider the graph of $y = f(x)$ as shown.



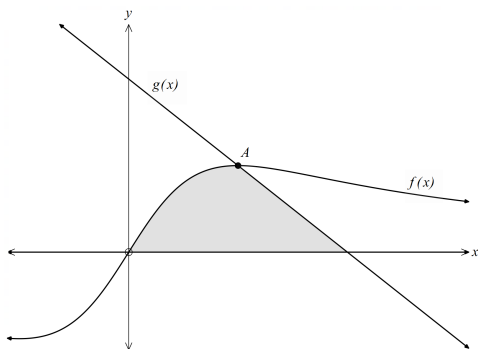
On the above number plane, draw a sketch of the function $y = 2f(x) + 6$, showing x and y intercepts and the coordinates of the point of inflection. 2

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Question 27 (4 marks)

The graph below shows the functions $f(x) = \frac{2x}{x^2+1}$ and $g(x) = -x + 2$.



- (i) Verify that the coordinates of A are $(1, 1)$. 1

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- (ii) Find the exact value of the shaded area in simplest form. 3

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Question 28 (5 marks)

Let $f(x) = x^4 + 2x^3 + 1$.

- (i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. 3

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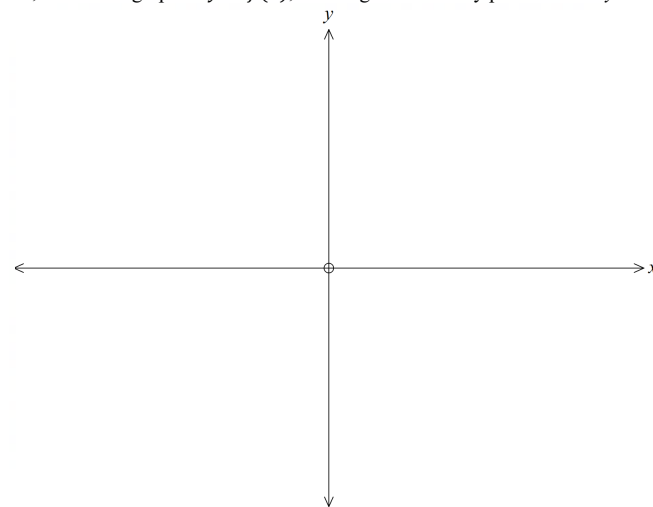
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- (ii) Hence, sketch the graph of $y = f(x)$, showing all stationary points and its y -intercept. 2



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Question 29 (6 marks)

The duration, in hours, that a suburban train trip takes is a random variable X , which is modelled by the probability density function:

$$f(x) = \begin{cases} \cos x, & 0 \leq x \leq k \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find the value of k . 2

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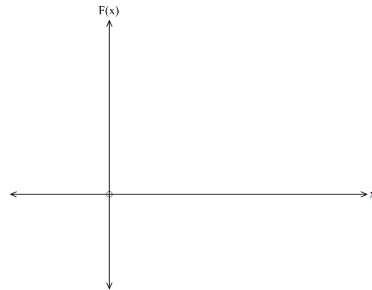
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- (ii) On the axes below, graph the cumulative density function. 2



- (iii) Find the probability that the train trip takes longer than one hour. 2

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Question 30 (4 marks)

In a local basketball competition, a team can either win or lose a match.

A certain team has a probability of 0.45 of winning a match if the coach's son is playing in that match or a probability of 0.75 if he is not. The probability of the coach's son playing is 0.4.

- (i) Find the probability that the team will win a match. 2

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- (ii) Given that the team has won a match, find the probability that the coach's son was playing on that day. 2

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Question 31 (3 marks)

The moon has a lower gravity than Earth. This means that a ball will bounce higher on the moon than Earth.

When a ball is dropped on the moon, each bounce is 85% as high as the previous bounce.
When an identical ball is dropped on Earth, each bounce is 40% as high as the previous bounce.

Two identical balls are dropped on the moon and on Earth, each from a height of 2 metres.

Calculate the difference in total vertical distance travelled by these balls. **3**

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Question 32 (3 marks)

Solve the equation $2 \ln(x + 4) - \ln x = \ln(2x + 2)$. **3**

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Question 33 (4 marks)

- (i) On the axes below, sketch the graphs of $y = 1 + \cos x$ and $y = \sin x$ for the domain $0 \leq x \leq 2\pi$. **2**



- (ii) Hence, or otherwise, solve $\sin^2 x + \cos^2 x + \cos x - \sin x = 0$, in the domain $0 \leq x \leq 2\pi$. **2**

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- (iii) For how many years will Elon's daughter be able to withdraw the full amount from the account? 2

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Question 35 (4 marks)

Let $f(x) = e^{-kx} + 4x$, where k is a positive, rational number.

- (i) Find, in terms of k , the x -coordinate of the stationary point of the graph $y = f(x)$. 3

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- (ii) State the values of k such that the x -coordinate of this stationary point is a positive number. 1

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Questions 36 and 37 on other side.

Question 36 (3 marks)

The velocity of a particle is given by $\dot{x} = 6t^2(t^3 + 2)$, where $\dot{x}$ is velocity in metres per second and t is time in seconds.

If its initial displacement is 4 metres, show that the particle is never at the origin.

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Question 37 (7 marks)

A closed rectangular prism has a height h cm, width x cm and length $1.5x$ cm. The surface area of the rectangular prism is 3240 cm^2 .

(i) Find an expression for h in terms of x .

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Question 37 continues on page 27

Year 12 Advanced Trial - Student Solutions

1. $2\cos x = \sqrt{3}$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$x = \frac{\pi}{6}, \frac{11\pi}{6} \quad \boxed{D}$$

2. Range of $y = 3^{-x}$ is $(0, \infty)$

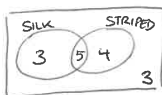
Range of $y = 3^{-x} + 2$ is $(2, \infty)$ $\boxed{C}$

3. $\int e + e^{4x} dx$

$$= ex + \frac{1}{4}e^{4x} + c \quad \boxed{B}$$

4. $15 - 3 = 12$

$8 + 9 - 12 = 5$ (silk and striped)



$P(\text{silk not striped}) = \frac{3}{15} = \frac{1}{5} \quad \boxed{C}$

5. P is a point of inflection so $f''(x) = 0$

Gradient at P is negative, so $f'(x) < 0$ $\boxed{A}$

6. $\frac{1}{5} + k + 2k + k + \frac{2}{5} = 1$

$$4k + \frac{3}{5} = 1$$

$$4k = \frac{2}{5}$$

$$k = \frac{1}{10} \quad \boxed{D}$$

7. $S_{\infty} = 18, a = 12, r = ?$

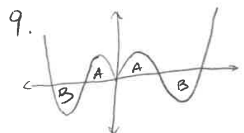
$$18 = \frac{12}{1-r}$$

$$1-r = \frac{12}{18}$$

$$r = 1 - \frac{12}{18}$$

$$r = \frac{1}{3} \quad \boxed{A}$$

8. Period = $\frac{2\pi}{n}$
 $= \frac{2\pi}{\frac{\pi}{5}}$
 $= 10 \quad \boxed{C}$



Since $g(x)$ is even, the graph is symmetrical about y-axis.

$$-B + A + A - B = -2$$

$$-B + 3 + 3 - B = -2$$

$$-2B = -8$$

$$B = 4$$

$$\text{Area} = 2 \times 4 + 2 \times 3$$

$$= 14 \quad \boxed{C}$$

10. $f'(x) = -2x + 4$

$$f(x) = -x^2 + 4x + c$$

$$-2x + 4 = 0$$

$$x = 2$$

so Max at $(2, 20)$

$$20 = -(2)^2 + 4(2) + c$$

$$c = 16$$

$$\therefore y = 16 + 4x - x^2 \quad \boxed{D}$$

Section II

11. $\frac{\sin x}{x}$

$$u = \sin x \quad v = x$$

$$u' = \cos x \quad v' = 1$$

$$\frac{dy}{dx} = \frac{x \cos x - 1 \sin x}{x^2}$$

$$= \frac{x \cos x - \sin x}{x^2}$$

12. $g(-2) = 2(-2) + 1$
 $= -3$

$$h(-3) = (-3)^2 - (-3)$$

$$= 12$$

b) $2(x^2 - x) + 1$

$$= 2x^2 - 2x + 1$$

c) $x^2 - x = 0$

$$x(x-1) = 0$$

$$x = 0, 1$$

13. $F = \frac{k}{L}$

$$9 = \frac{k}{14}$$

$$k = 9 \times 14$$

$$= 126$$

$$\therefore F = \frac{126}{L}$$

14. $S_{14} = 2023$

$$n = 14$$

$$a = 34$$

$$d = ?$$

$$2023 = \frac{14}{2}(2 \times 34 + (14-1)d)$$

$$2023 = 7(68 + 13d)$$

$$289 = 68 + 13d$$

$$221 = 13d$$

$$d = 17$$

15. $\int_0^{\frac{\pi}{2}} \sec^2 2x dx$

$$= \left[\frac{1}{2} \tan(2x) \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \tan(2 \times \frac{\pi}{2}) - \frac{1}{2} \tan(2 \times 0)$$

$$= \frac{1}{2} \times \sqrt{3} - \frac{1}{2} \times 0$$

$$= \frac{\sqrt{3}}{2}$$

16. $y = 2(4x-3)^4$

$$\frac{dy}{dx} = 8(4x-3)^3 \times 4$$

$$= 32(4x-3)^3$$

Let $x = 1$

$$M_{\text{tangent}} = 32(4(1)-3)^3$$

$$= 32$$

$$M_{\text{normal}} = -\frac{1}{32}$$

Let $x = 1$

$$y = 2(4(1)-3)^4$$

$$y = 2$$

$$y - 2 = -\frac{1}{32}(x-1)$$

$$32y - 64 = -x + 1$$

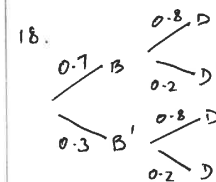
$$x + 32y - 65 = 0$$

17. $\int_1^7 \frac{8}{x+1} dx$

$$\frac{x}{y} \begin{array}{c} 1 \ 3 \ 5 \ 7 \\ 4 \ 2 \ \frac{4}{3} \ 1 \end{array}$$

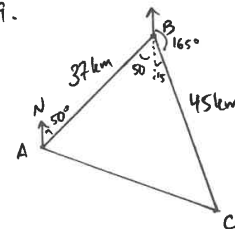
$$\approx \frac{2}{2} \left[4 + 1 + 2 \left(2 + \frac{4}{3} \right) \right]$$

$$\approx \frac{35}{3}$$



$P(\text{at least one passing}) = 1 - P(\text{both failing})$
 $= 1 - 0.3 \times 0.2$
 $= 0.94$

19.



i) $50 + 15 = 65^\circ$

ii) $AC^2 = 37^2 + 45^2 - 2 \times 37 \times 45 \times \cos 65$

$$AC^2 = 1986.62 \dots$$

$$AC = 44.6 \text{ km}$$

20. i) $E(x) = 0 \times 0.1 + 1 \times 0.35 + 2 \times 0.4 + 3 \times 0.15$
 $= 1.6$

ii) $24 \div 1.6$
 $= 15 \text{ visits}$

$$21. i) P = Ae^{-kt}$$

$$\frac{dP}{dt} = -kAe^{-kt}$$

$$= -kP$$

$$ii) A = 5000$$

$$4250 = 5000e^{-5k}$$

$$\frac{17}{20} = e^{-5k}$$

$$\ln\left(\frac{17}{20}\right) = -5k$$

$$k = 0.0325$$

$$iii) \frac{dP}{dt} = -0.0325 \times 5000e^{-0.0325 \times 10}$$

$$= -117.4$$

$$22. x^2 = 25$$

$$x = \pm 5$$

since $x < 0$ for $f(x) = x^2$
 $\therefore x = -5$ only

$$2x - 3 = 25$$

$$2x = 28$$

$$x = 14$$

since $x > 0$ for $f(x) = 2x - 3$
then $x = 14$ is included
 $\therefore x = -5, 14$

$$23. b(x+a)^2 = 2x^2 + 16x + 35$$

$$b = 2$$

$$2(x^2 + 2ax + a^2) + 3 = 2x^2 + 16x + 35$$

$$2x^2 + 4ax + 2a^2 + 3 = 2x^2 + 16x + 35$$

$$2a^2 + 3 = 35 \quad 4ax = 16$$

$$2a^2 = 32 \quad 4a = 16$$

$$a^2 = 16 \quad a = 4$$

$$a = 4 \text{ satisfies both.}$$

$$24. f(x) = \frac{4}{81}(4x^2 - x^3) \quad 1 \leq x \leq 4$$

$$f'(x) = \frac{4}{81}(8x - 3x^2)$$

$$0 = \frac{4}{81}(8x - 3x^2)$$

$$0 = x(8 - 3x)$$

$$x = 0 \text{ (not in domain)} \quad x = \frac{8}{3}$$

$$f''(x) = \frac{4}{81}(8 - 6x)$$

$$f''\left(\frac{8}{3}\right) = -\frac{32}{81} < 0 \therefore \text{MAX}$$

$$\therefore \text{mode is } x = \frac{8}{3}$$

$$25. i) \frac{d}{dx} 2x \ln x$$

$$u = 2x \quad v = \ln x$$

$$u' = 2 \quad v' = \frac{1}{x}$$

$$\frac{dy}{dx} = 2 \ln x + 2$$

$$ii) \int 1 + \ln x \, dx$$

$$= \frac{1}{2} \int 2 + 2 \ln x \, dx$$

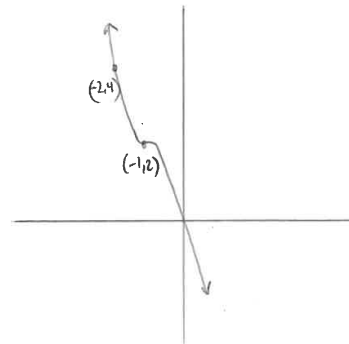
$$= \frac{1}{2} (2x \ln x + C)$$

$$x \ln x + C$$

$$26. y = 2f(x) + 6$$

y-coordinates $\times 2$ then $+6$

(0, -3) (0, -6) (0, 0)
(-1, -2) (-1, -4) (-1, 2)



$$27. i) f(1) = \frac{2(1)}{1^2 + 1} = 1$$

$$g(1) = -1 + 2 = 1$$

$$\therefore (1, 1) \text{ satisfies both equations.}$$

$$ii) \int_0^{\frac{\pi}{2}} \frac{2x}{x^2 + 1} \, dx$$

$$= [\ln|x^2 + 1|]_0^{\frac{\pi}{2}}$$

$$= \ln|1^2 + 1| - \ln|0^2 + 1|$$

$$= \ln 2$$

$$\therefore \text{Area} = \frac{1}{2} + \ln 2$$

$$28. i) f(x) = x^4 + 2x^3 + 1$$

$$f'(x) = 4x^3 + 6x^2 \quad f''(x) = 12x^2 + 12x$$

$$0 = 4x^3 + 6x^2$$

$$0 = 2x^2(2x + 3)$$

$$x = 0, x = -\frac{3}{2}$$

$$f(0) = 1 \quad f\left(-\frac{3}{2}\right) = \left(-\frac{3}{2}\right)^4 + 2\left(-\frac{3}{2}\right)^3 + 1$$

$$= -\frac{11}{16}$$

$$f''\left(-\frac{3}{2}\right) = 12\left(-\frac{3}{2}\right)^2 + 12\left(-\frac{3}{2}\right)$$

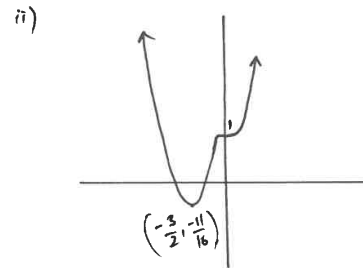
$$= 9 > 0 \therefore \text{min}$$

$$\left(-\frac{3}{2}, -\frac{11}{16}\right) \text{ is a minimum}$$

$$f''(0) = 12(0)^2 + 12(0)$$

$$= 0 \therefore \text{point of inflection}$$

x	-1	0	1
$f(x)$	2	0	10

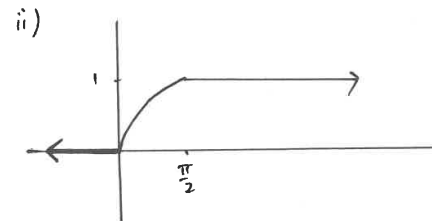


$$29. i) \int_0^{\frac{\pi}{2}} \cos x \, dx = 1$$

$$[\sin x]_0^{\frac{\pi}{2}} = 1$$

$$\sin \frac{\pi}{2} - \sin 0 = 1$$

$$k = \frac{\pi}{2}$$



$$iii) P(X > 1) = 1 - P(X < 1)$$

$$= 1 - \sin 1$$

$$= 0.159$$

$$30. i)$$

0.4 S 0.45 W
0.55 W'

0.6 S 0.75 W
0.25 W'

$$= 0.4 \times 0.45 + 0.6 \times 0.75$$

$$= 0.63$$

$$ii) P(S|W) = \frac{P(S \cap W)}{P(W)}$$

$$= \frac{0.4 \times 0.45}{0.63}$$

$$= \frac{2}{7}$$

Mountain	Earth
2m ↓ 1.7 ↓ 1.445 ↓ ...	2m ↓ 0.8 ↓ 0.32 ↓
Total distance Mountain	Total distance Earth
$= 2 + 2(1.7 + 1.445 + \dots)$	$= 2 + 2(0.8 + 0.32 + \dots)$
$= 2 + 2\left(\frac{1.7}{1 - 0.85}\right)$	$= 2 + 2\left(\frac{0.8}{1 - 0.4}\right)$
$= \frac{74}{3}$	$= \frac{14}{3}$
$\frac{74}{3} - \frac{14}{3} = 20m$	

$$32. 2 \ln(x+4) - \ln x = \ln(2x+2)$$

$$\ln(x+4)^2 - \ln x = \ln(2x+2)$$

$$\ln\left(\frac{(x+4)^2}{x}\right) = \ln(2x+2)$$

$$\frac{(x+4)^2}{x} = 2x+2$$

$$x^2 + 8x + 16 = 2x^2 + 2x$$

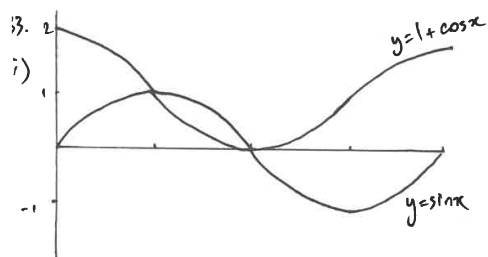
$$x^2 - 6x - 16 = 0$$

$$(x-8)(x+2) = 0$$

$$x=8, x=-2$$

↳ does not satisfy equation

$$\therefore x=8 \text{ only}$$



$$ii) \sin^2 x + \cos^2 x + \cos x - \sin x = 0$$

$$1 + \cos x = \sin x$$

$\therefore$ points of intersection

$$x = \frac{\pi}{2}, \pi$$

$$34. i) 8\% \div 2 = 4\%$$

$$= 30000(1.04)^2 - 4080$$

$$= \$28368$$

$$ii) A_1 = 30000(1.04)^2 - 4080$$

$$A_2 = A_1(1.04)^2 - 4080$$

$$= 30000(1.04)^4 - 4080(1.04)^2 - 4080$$

$$A_3 = A_2(1.04)^2 - 4080$$

$$= 30000(1.04)^6 - 4080(1.04)^4 - 4080(1.04)^2 - 4080$$

$\vdots$

$$A_n = 30000(1.04)^{2n} - 4080(1 + 1.04^2 + 1.04^4 + \dots)$$

$$= 30000(1.04)^{2n} - 4080\left(\frac{1(1.04)^{2n} - 1}{1.04^2 - 1}\right)$$

$$= 30000(1.04)^{2n} - 50000(1.04^{2n} - 1)$$

$$= 50000 - 20000(1.04)^{2n}$$

$$iii) A_n = 0$$

$$0 = 50000 - 20000(1.04)^{2n}$$

$$50000 = 20000(1.04)^{2n}$$

$$\frac{5}{2} = 1.04^{2n}$$

$$\ln\left(\frac{5}{2}\right) = \ln(1.04^{2n})$$

$$\ln\left(\frac{5}{2}\right) = 2n \ln(1.04)$$

$$2n = \frac{\ln\left(\frac{5}{2}\right)}{\ln(1.04)}$$

$$= 23.36 \dots$$

$$n = 11.68 \dots$$

$$\therefore 11 \text{ years.}$$

$$35 i) f(x) = e^{-kx} + 4x$$

$$f'(x) = -ke^{-kx} + 4$$

$$0 = -ke^{-kx} + 4$$

$$ke^{-kx} = 4$$

$$e^{-kx} = \frac{4}{k}$$

$$-kx = \ln\left(\frac{4}{k}\right)$$

$$x = -\frac{1}{k} \ln\left(\frac{4}{k}\right)$$

$$x = \frac{1}{k} \ln\left(\frac{k}{4}\right)$$

$$ii) \frac{k}{4} > 1$$

$$k > 4$$

$$36. x = \int 6t^2(t^3+2) dt \quad \text{or} \quad x = \int 6t^5 + 12t^2 dt$$

$$x = (t^3+2)^2 + c$$

$$\text{let } t=0, x=4$$

$$4 = (0^3+2)^2 + c$$

$$c=0$$

$$\therefore x = (t^3+2)^2$$

$$\text{let } x=0$$

$$0 = (t^3+2)^2$$

$$-2 = t^3$$

$$\sqrt[3]{-2} = t$$

$$t > 0$$

$\therefore x$ can never equal zero
so particle never at the origin

$$37.$$



$$SA = 3240$$

$$i) SA = 2(x \times 1.5x + xh + 1.5xh)$$

$$= 3x^2 + 5hx$$

$$3240 = 3x^2 + 5hx$$

$$3240 - 3x^2 = 5hx$$

$$h = \frac{3240 - 3x^2}{5x}$$

$$h = \frac{648}{x} - \frac{3x}{5}$$

$$ii) V = 1.5x^2h$$

$$V = 1.5x^2\left(\frac{648}{x} - \frac{3x}{5}\right)$$

$$V = 972x - \frac{9}{10}x^3$$

$$\frac{dV}{dx} = 972 - \frac{27}{10}x^2$$

$$0 = 972 - \frac{27}{10}x^2$$

$$\frac{27}{10}x^2 = 972$$

$$x^2 = 360$$

$$x = \pm\sqrt{360}$$

$$x = 6\sqrt{10} \quad (x > 0)$$

$$\frac{d^2V}{dx^2} = -\frac{54}{10}x$$

$$\text{let } x = 6\sqrt{10}$$

$$\frac{d^2V}{dx^2} = -\frac{162\sqrt{10}}{5} < 0 \therefore \text{MAX}$$

$$h = \frac{648}{6\sqrt{10}} - \frac{3 \times 6\sqrt{10}}{5}$$

$$= \frac{108}{\sqrt{10}} - \frac{18\sqrt{10}}{5}$$

$$= \frac{108\sqrt{10}}{10} - \frac{18\sqrt{10}}{5}$$

$$= \frac{54\sqrt{10}}{5} - \frac{18\sqrt{10}}{5}$$

$$= \frac{36\sqrt{10}}{5}$$

$$x = 6\sqrt{10} \quad h = \frac{36\sqrt{10}}{5}$$